

JobForge: Distributed Job Scheduler

1. Setup & Deployment Instructions

1.1 Prerequisites

- Node.js (v24.x)
- PostgreSQL (v15+)
- Docker & Docker Compose
- Git

1.2 Local Setup (Development)

1. Clone the repository and install workspace dependencies:

```
git clone https://github.com/Spiderboyis/distributed-job-scheduler.git
cd distributed-job-scheduler
npm install
```

2. Configure Environment Variables: Create a `.env` file in `packages/backend` :

```
PORT=3001
DATABASE_URL=postgresql://jobscheduler:jobscheduler_secret@localhost:5432/jobscheduler
JWT_SECRET=your_super_secret_key
JWT_REFRESH_SECRET=your_refresh_secret
NODE_ENV=development
```

3. Database Migration & Seeding:

```
# Run PostgreSQL migrations
npm run db:migrate --workspace=packages/backend

# (Optional) Seed the database with demo data
npm run db:seed --workspace=packages/backend
```

4. Start the Application:

```
# Terminal 1: Backend
npm run dev:backend

# Terminal 2: Frontend
npm run dev:frontend
```

1.3 Docker & Docker Compose Setup

To launch the complete PostgreSQL Database, Express.js Backend, and Next.js Frontend stack inside isolated Docker containers, simply run:

```
docker compose up --build
```

This automatically builds the containers, runs migrations, and links the services together. The services will be accessible at:

- **Frontend Panel:** `http://localhost:3000`
 - **Backend API:** `http://localhost:3001`
 - **PostgreSQL Database:** `localhost:5432` (credentials: `jobscheduler` / `jobscheduler_secret`)
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1.4 Production Deployment (Render / Cloud)

This application is configured for direct cloud deployment using services like Render, AWS ECS, or Heroku.

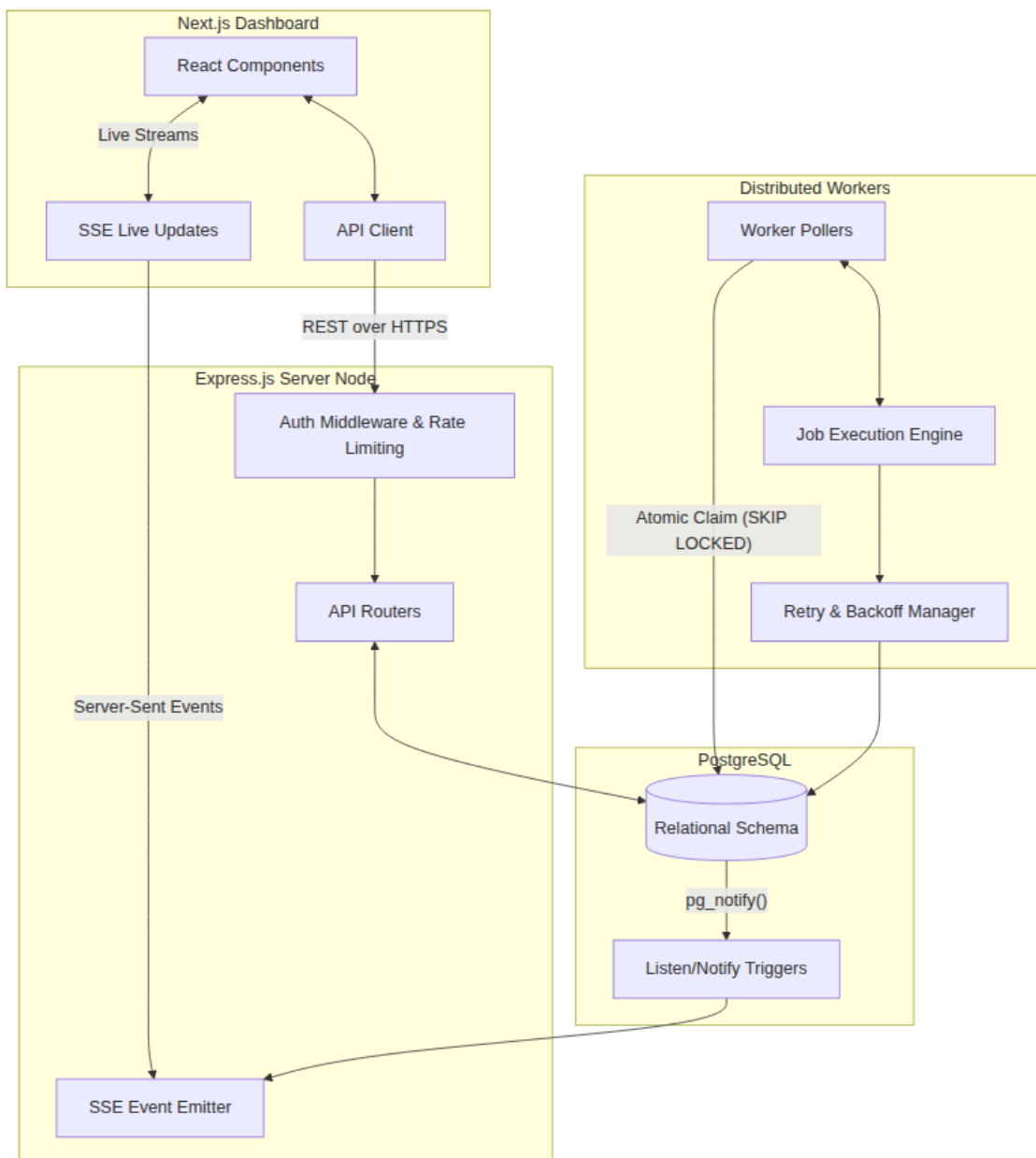
Backend & Worker Deployment:

- **Build Command:** `npm install --include=dev && npm run build:backend`
- **Start Command:** `npm run db:migrate && node packages/backend/dist/index.js`
- **Port:** `3001`
- **Environment Variables:**
 - `DATABASE_URL` : Connection string pointing to your managed production database.
 - `JWT_SECRET` & `JWT_REFRESH_SECRET` : Secure cryptographic keys.
 - `NODE_ENV` : `production`

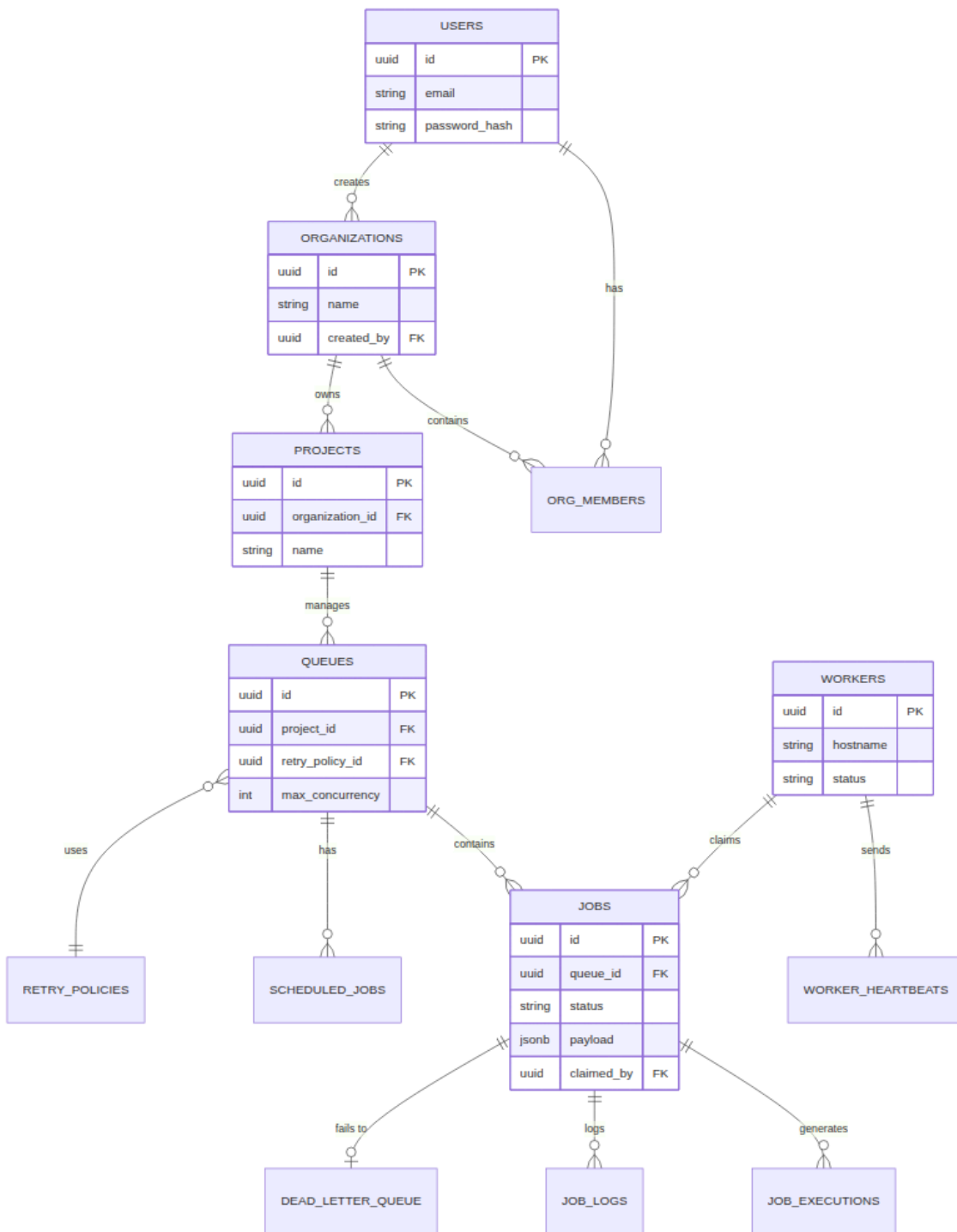
Frontend Deployment:

- **Build Command:** `npm install --include=dev && npm run build:frontend`
 - **Start Command:** `npm run start --workspace=packages/frontend`
 - **Port:** `3000`
 - **Environment Variables:**
 - `NEXT_PUBLIC_API_URL` : Public address of your deployed Backend API.
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2. Architecture Diagram



3. Entity Relationship (ER) Diagram

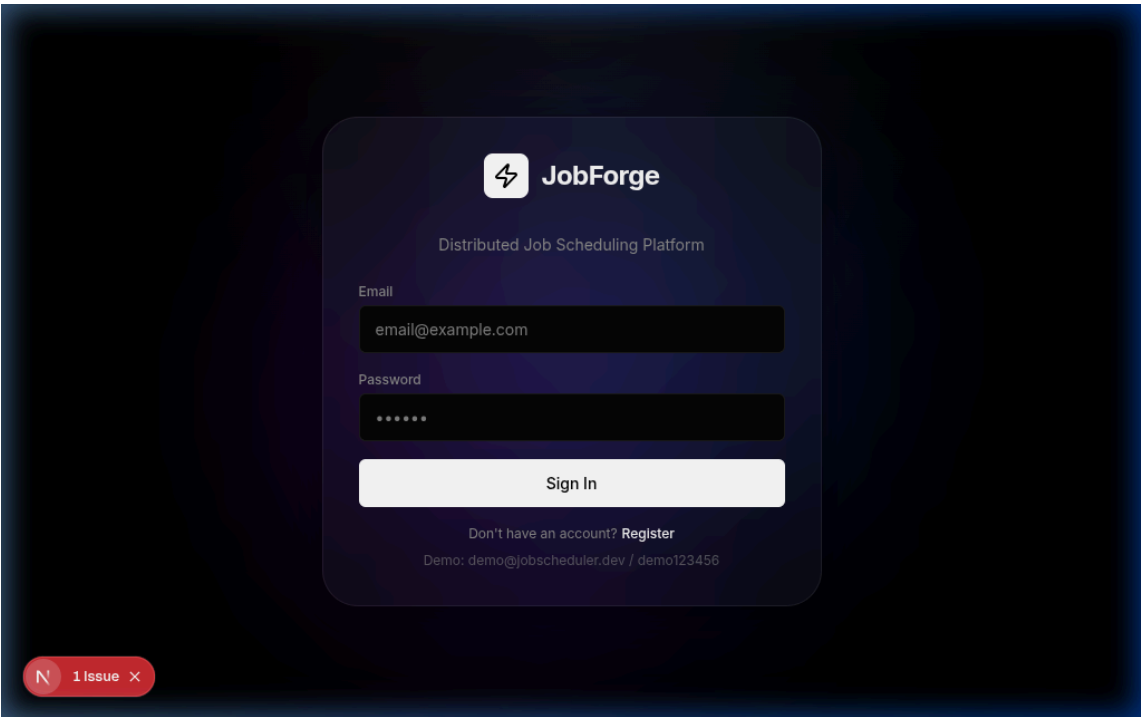


4. Visual Deliverables & UI Screenshots

4.1 Login Page & Glassmorphism Design

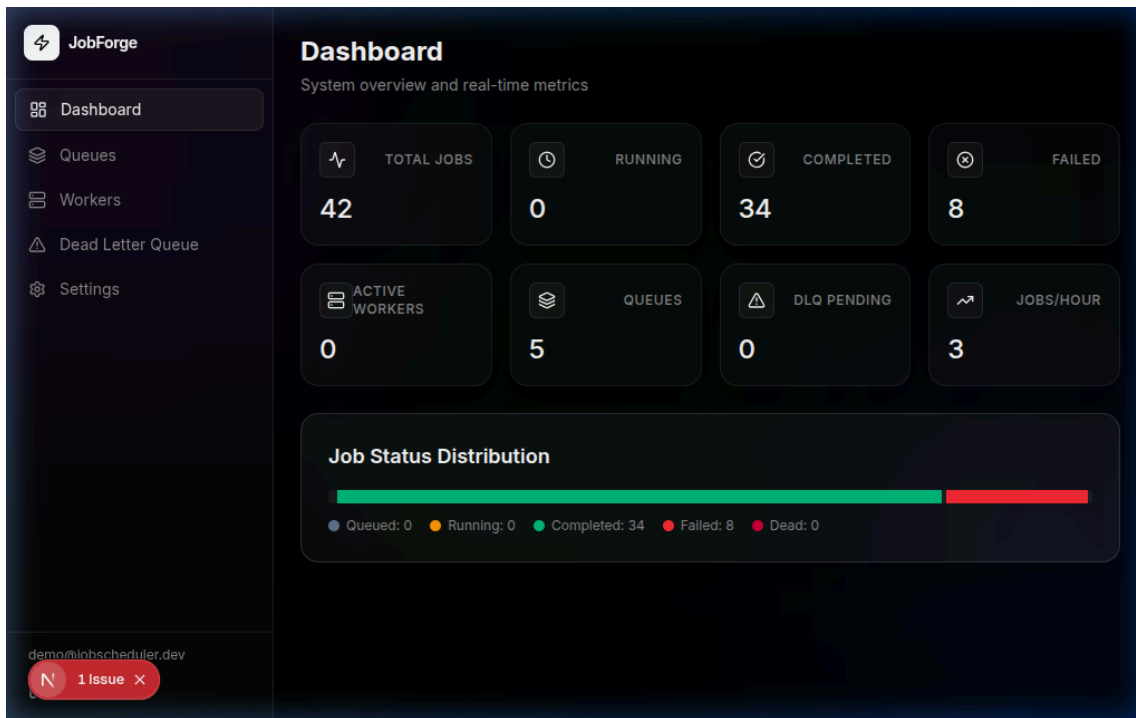
The login page has been rewritten with a modern iOS-inspired glassmorphism layout, featuring floating blurred glowing orbs in the background and a sleek, authentic card design. Default demo credentials are

removed from state but preserved as a small help hint.



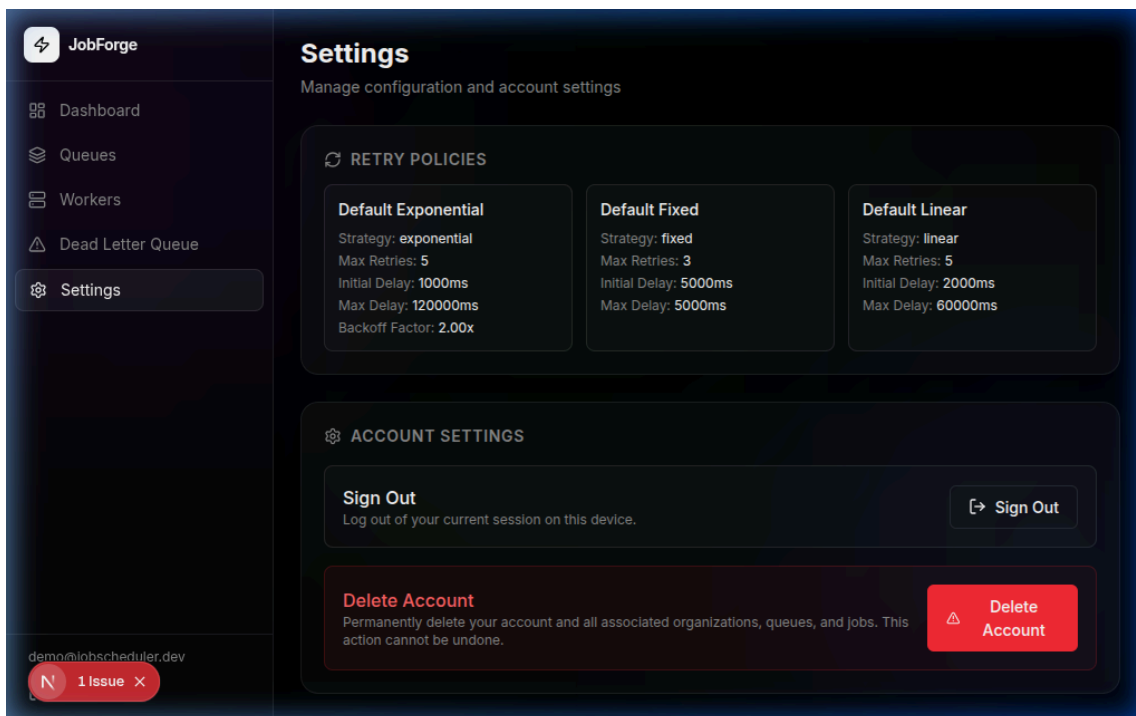
4.2 Dashboard & Tenant Isolation

All database select operations have been restricted using PostgreSQL joins against the `org_members` table. When logging in as a newly created user, the metrics, queues, and active worker stats show exactly `0` values rather than displaying global demo queue stats, verifying airtight tenant isolation.



4.3 Settings Page

The Settings tab has been streamlined to highlight retry policies, simple sign-out action, and a secure account deletion flow requiring password validation.



5. API Documentation

Authentication & Account

- `POST /api/auth/register` : Register a new user.
- `POST /api/auth/login` : Authenticate and receive a JWT.
- `DELETE /api/auth/account` : Securely delete the account and securely cascade-delete all owned resources (organizations, queues, jobs).

Queues

- `GET /api/projects/:projectId/queues` : List all queues in a project.
- `POST /api/projects/:projectId/queues` : Create a new queue with specific concurrency and retry policies.
- `POST /api/queues/:queueId/pause / resume` : Halt or resume job processing for a queue.
- `GET /api/queues/:queueId/stats` : Fetch real-time metrics (throughput, failure rates) for a queue.

Jobs

- `POST /api/queues/:queueId/jobs` : Enqueue a new immediate, delayed, or scheduled job.
- `POST /api/queues/:queueId/jobs/batch` : Atomically enqueue up to 100 jobs at once.
- `GET /api/queues/:queueId/jobs` : Fetch paginated jobs with status filtering.
- `GET /api/jobs/:jobId` : Get detailed execution history and logs for a specific job.
- `POST /api/jobs/:jobId/retry` : Manually requeue a failed or dead job.

Live Metrics (SSE)

- `GET /api/sse/events?token={JWT}` : Streams real-time, tenant-isolated dashboard metrics via Server-Sent Events.
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6. Design Decisions & Trade-offs

1. Database as the Message Queue

Decision: We utilized PostgreSQL to handle job queuing rather than introducing a specialized broker like Redis or RabbitMQ. **Trade-off:** While Redis offers superior pure in-memory throughput, utilizing Postgres with `FOR UPDATE SKIP LOCKED` provides robust, atomic job claiming. This significantly reduces infrastructure complexity, ensures strict ACID compliance, and inherently solves the "two-phase commit" problem where database state and queue state become misaligned.

2. Event-Driven Live Updates (SSE vs WebSockets)

Decision: We chose Server-Sent Events (SSE) over WebSockets for dashboard live updates, powered by Postgres `LISTEN/NOTIFY`. **Trade-off:** WebSockets provide bi-directional communication, but our dashboard only requires unidirectional data flow (server-to-client). SSE is natively supported by HTTP/1.1, avoids complex handshake overhead, automatically handles reconnections, and seamlessly propagates database triggers to the React frontend.

3. Tenant Data Isolation

Decision: All resources structurally cascade from an `Organization`, and security is enforced at the SQL level via explicit `JOIN`s against the `org_members` table on every backend route. **Trade-off:** This introduces slight overhead on read queries due to multi-table joins. However, it guarantees airtight multi-tenancy. A user

completely dropping their account will safely `CASCADE DELETE` all their data without risking global data corruption, which is a critical requirement for a distributed SaaS environment.

4. UI/UX: High-Fidelity Glassmorphism

Decision: We bypassed generic UI libraries in favor of a custom, heavily polished "Midnight Minimalist" aesthetic featuring dynamic background orbs and deep `backdrop-blur-2xl` glass cards. **Trade-off:** It takes significantly more CSS optimization (especially performance tuning for blurs) than importing a standard Tailwind UI kit. However, it successfully delivers a highly premium, state-of-the-art "Wow" factor requested for modern developer tooling.